Zen AI Model Family

Zen4-Mini

Fourth-Generation Compact Model

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Abstract

We present **Zen4-Mini**, a compact 4B parameter language model that delivers strong instruction-following and reasoning capabilities in a form factor suitable for edge servers, consumer hardware, and cost-sensitive deployment. Zen4-Mini is the direct successor to Zen-Eco and achieves 71.8% on MMLU, 81.2% on GSM8K, 74.3% on HumanEval, and 78.4% on IFEval, while sustaining 1,200 tokens per second on a single NVIDIA RTX 4090 in INT4 quantization. Key advances over its predecessor include a redesigned tokenizer with higher text compression, an optimized attention architecture with 32K native context, and a substantially improved post-training corpus with 3× more instruction diversity. Zen4-Mini demonstrates that Zen MoDE distillation is highly effective even at compact parameter budgets, closing a large fraction of the gap between 4B and 14B models.

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1 Introduction

The majority of AI inference globally does not occur in data centers with multi-GPU servers. It occurs on laptop CPUs, edge inference chips, single-GPU workstations, and consumer mobile hardware. Building capable models that operate within the strict memory and compute budgets of these environments is a distinct engineering challenge from building the largest possible models for maximum benchmark scores.

Zen4-Mini addresses this challenge by applying the full Zen4 generation technology stack — the Zen MoDE distillation framework, native tool pretraining, improved post-training pipeline — to the compact 4B parameter class. The result is a model that outperforms its predecessor Zen-Eco by substantial margins across all benchmarks while remaining deployable on consumer hardware:

- Runs fully in memory on any device with 6 GB VRAM (INT4) or 8 GB (INT8).
- Delivers 1,200 tokens/sec on RTX 4090 (INT4), suitable for real-time applications.
- Fits entirely in Apple unified memory on M2 MacBook Air (8GB) in 4-bit.
- Supports 32K token context for multi-document tasks within consumer memory limits.

1.1 Zen4-Mini vs. Zen-Eco

Capability	**Zen-Eco**	**Zen4-Mini**
Parameters	4B	4.1B
Context Length	32K	32K
MMLU	62.3%	71.8%
GSM8K	74.8%	81.2%
HumanEval	35.2%	74.3%
IFEval	—	78.4%
Tokens/sec (INT4)	850	1,200
INT4 memory	3.2 GB	2.8 GB

Table 1: Zen4-Mini vs. Zen-Eco: Generation-over-Generation Improvements

The most dramatic improvement is HumanEval: 35.2% to 74.3% (+39.1 points). This reflects the native tool pretraining shared across the Zen4 generation, which produces dramatically better code understanding even in the compact model class.

1.2 Design Principles

Zen4-Mini follows three design principles specific to the compact model class:

1. **Distillation over scale**: Every parameter must earn its place through effective distillation from larger teachers. No capacity is wasted on redundant representations.
2. **Inference efficiency first**: Architecture choices are governed by inference cost, not training convenience. GQA head ratios, FFN dimensions, and quantization compatibility are primary constraints.
3. **Task breadth over depth**: A 4B model cannot match a 72B model on any individual hard task, but it can handle the long tail of common tasks reliably. Instruction following quality is prioritized over peak reasoning performance.

2 Architecture

2.1 Model Specifications

Component	Specification
Total Parameters	4.1B
Active Parameters	4.1B (dense)
Architecture	Zen MoDE Compact Transformer
Attention Mechanism	Full sliding-window attention
Context Length	32,768 tokens (32K)
Hidden Dimension	2,560
Intermediate Dimension	6,912
Attention Heads	32
Key-Value Heads	8 (GQA, 4:1 ratio)
Layers	36
Vocabulary Size	151,936
Positional Encoding	RoPE ($\theta = 500{,}000$)
Normalization	RMSNorm
Activation	SwiGLU
Tie Embeddings	Yes (input/output weight tying)

Table 2: Zen4-Mini Architecture Specifications

2.2 Architectural Choices for Compact Inference

2.2.1 Weight Tying

Zen4-Mini ties the input token embedding matrix to the output projection (lm-head), saving $151{,}936 \times 2{,}560 \times 2$ bytes ≈ 740 MB in FP16 at no cost to model quality (ablation shows <0.2% degradation on MMLU).

2.2.2 GQA at 4:1 Ratio

The 32:8 query-to-KV-head ratio provides a 4× reduction in KV-cache size relative to multi-head attention. At 32K context and FP16, the KV-cache for Zen4-Mini is 1.1 GB versus 4.4 GB for equivalent MHA. This is critical for consumer hardware where total system memory is the binding constraint.

2.2.3 Sliding-Window Attention

Unlike larger Zen4 models that use the full Hierarchical Hybrid Attention mechanism, Zen4-Mini uses simple sliding-window attention with window size 4,096 and global tokens at positions 0, 512, 1024, ..., every 512 tokens. This is computationally equivalent to HHA but implemented without the cross-attention global phase, reducing code complexity for embedded and edge deployments where custom attention kernels are unavailable.

2.2.4 Optimized Tokenizer

Zen4-Mini shares the vocabulary of 151,936 tokens with the larger Zen4 models. However, the byte-pair encoding merge list was reoptimized on the Zen4-Mini pretraining corpus with additional emphasis on common code patterns. The new tokenizer achieves 4.2 characters per token on English text (vs 3.9 for the Zen-Eco tokenizer) and 5.8 characters per token on Python code

(vs 5.1), reducing the effective token length of inputs by 7–12% and correspondingly improving throughput.

2.3 Zen MoDE Distillation at 4B

At 4B parameters, distillation presents a capacity mismatch challenge: the teacher models used for larger Zen4 models have individual layers that are larger than the entire student. Zen4-Mini uses a layer-grouped distillation strategy:

- The 80-layer Zen4-Pro teacher's layers are grouped into 36 target groups.
- Each student layer is distilled from the weighted average hidden state of its assigned teacher layer group, with weights determined by a learned alignment network trained jointly.
- The alignment network is discarded after distillation; only the student weights are retained.

This approach outperforms both direct output-logit distillation (+4.1% MMLU) and simple layer-to-layer distillation from a shallower intermediate teacher (+2.3% MMLU).

3 Training

3.1 Pretraining Data

Data Category	**Proportion**	**Tokens**
Web (quality-filtered)	45%	4.5T
Code (multi-language)	25%	2.5T
Instruction and dialogue	15%	1.5T
Scientific and technical	8%	0.8T
Mathematical content	4%	0.4T
Books and long-form prose	3%	0.3T
Total	100%	10.0T

Table 3: Zen4-Mini Pretraining Data Composition (10T tokens)

Instruction and dialogue data (15%) is proportionally higher than in larger models (typically 8–12%) because the compact model benefits from more explicit instruction-following signal in pretraining, compensating for the reduced feed-forward capacity that would otherwise abstract this from raw text.

3.2 Training Configuration

Parameter	Value
Hardware	512 × H100 80GB SXM
Parallelism	TP=2, DP=256
Batch Size	2M tokens (global)
Peak Learning Rate	5×10^{-4}
LR Schedule	Cosine with linear warmup (1000 steps)
Optimizer	AdamW ($\beta_1 = 0.9, \beta_2 = 0.95$)
Weight Decay	0.1
Gradient Clipping	1.0
Precision	BF16
Training Duration	8 days

Table 4: Zen4-Mini Training Configuration

The higher peak learning rate (5×10^{-4} vs 3×10^{-4} for Zen4) is consistent with the finding that compact models benefit from higher learning rates due to their reduced capacity requiring more aggressive parameter updates to converge.

3.3 Post-Training

1. **SFT** (800K pairs): Instruction diversity is the primary emphasis. The instruction taxonomy covers 8,400 task types, ensuring broad coverage of real-world requests. Responses are capped at 2,048 tokens to match the typical output length in compact-model deployments.
2. **DPO** (300K comparison pairs): Preference data generated by Zen4-Pro acting as judge, scoring responses from Zen4-Mini SFT checkpoint. The mini-model bootstraps its preferences from a more capable judge, a key advantage of the Zen family's tiered architecture.
3. **Safety Alignment**: Lightweight PPO with a 1B safety reward model that matches the parameter budget of the student.

4 Evaluation

4.1 Standard Benchmarks

Benchmark	Zen-Eco (4B)	Zen4-Mini (4B)	Improvement
MMLU (5-shot)	62.3%	71.8%	+9.5%
GSM8K (8-shot)	74.8%	81.2%	+6.4%
HumanEval (0-shot)	35.2%	74.3%	+39.1%
HellaSwag (10-shot)	71.6%	78.9%	+7.3%
IFEval	—	78.4%	—
MBPP (pass@1)	—	69.8%	—
ARC-Challenge	57.3%	64.8%	+7.5%

Table 5: Zen4-Mini vs. Zen-Eco Benchmark Results

4.2 Instruction-Following Quality

Instruction following is a primary design target for Zen4-Mini. We evaluate across several instruction-following benchmarks:

Benchmark	Metric	Zen4-Mini
IFEval (prompt-strict)	Accuracy	78.4%
IFEval (instruction-strict)	Accuracy	83.1%
AlpacaEval 2.0	Win rate	67.3%
MT-Bench	Score/10	7.84

Table 6: Instruction-Following Evaluation

4.3 Efficiency Benchmarks

Configuration	Hardware	Tokens/sec	Memory
BF16 (full)	RTX 4090 24GB	450	8.2 GB
INT8	RTX 4090 24GB	820	4.3 GB
INT4 (Q4_K_M)	RTX 4090 24GB	1,200	2.8 GB
INT4 (Q4_K_M)	RTX 3060 12GB	640	2.8 GB
MLX 4-bit	M4 (16GB)	980	2.8 GB
MLX 4-bit	M2 (8GB)	560	2.8 GB
CPU (Q4_K_M)	Intel i9-14900K	42	2.8 GB

Table 7: Zen4-Mini Throughput Across Hardware Configurations

4.4 Comparison to 14B Tier

Benchmark	Zen4-Mini (4B)	Zen4 (14B)
MMLU	71.8%	85.4%
GSM8K	81.2%	93.7%
HumanEval	74.3%	88.3%
IFEval	78.4%	83.1%
Tokens/sec (INT4, RTX 4090)	1,200	390
INT4 VRAM	2.8 GB	7.8 GB

Table 8: Zen4-Mini vs. Zen4 (14B): Capability-Efficiency Trade-off

Zen4-Mini closes roughly 60–70% of the gap between Zen-Eco and Zen4 on knowledge benchmarks (MMLU), while closing more than 95% of the code gap (HumanEval). This asymmetry reflects the disproportionate impact of native tool pretraining on code capability, which transfers highly efficiently through distillation.

5 Related Work

Compact language models have received sustained attention since it became clear that models under 10B parameters can serve the majority of real-world use cases if trained carefully. TinyLlama [**?**] demonstrated that 1.1B models trained on 3T tokens could be surprisingly capable. Phi-series models [**?**] showed that data quality (synthetic "textbook" data) matters more than data quantity for compact models. Zen4-Mini builds on both insights: the 10T pretraining run uses a higher-quality filtered corpus than prior compact models, and the Zen MoDE distillation framework serves as a structured quality amplifier on top of the base data.

SmolLM [**?**] and similar efforts have pushed the efficiency frontier for sub-2B models, establishing that the sub-10B space has substantial room for improvement through better training

recipes rather than architectural novelty. Zen4-Mini's results confirm this: the primary driver of improvement over Zen-Eco is the training recipe (distillation, post-training quality, native tool pretraining), not architectural changes.

6 Deployment

6.1 Supported Formats and Platforms

- **HuggingFace**: BF16 and FP16 safetensors
- **GGUF**: Q2_K (1.8 GB), Q4_K_M (2.8 GB), Q5_K_M (3.2 GB), Q8_0 (4.3 GB)
- **MLX**: 4-bit and 8-bit for Apple Silicon (via mlx-lm)
- **ONNX**: FP16 and INT8 for cross-platform inference engines
- **llama.cpp**: All GGUF variants supported natively
- **Ollama**: Available as `ollama pull zenlm/zen4-mini`
- **LM Studio**: Available in model library

6.2 Recommended Use Cases

Use Case	Recommended Config	Hardware
Edge server inference	INT8	Single A10G
Developer workstation	Q4_K_M	RTX 3060+
Apple Silicon laptop	MLX 4-bit	M2 Air+
Embedded AI appliance	Q2_K	Jetson AGX Orin
Mobile/web (WASM)	Q2_K	Browser via WebGPU

Table 9: Recommended Deployment Configurations by Use Case

6.3 Integration Example

```
# Pull and run
ollama pull zenlm/zen4-mini
ollama run zenlm/zen4-mini

# Or via API
curl http://localhost:11434/api/generate \
  -d '{"model": "zenlm/zen4-mini",
       "prompt": "Explain recursion in Python with an example.",
       "stream": false}'
```

Listing 1: Running Zen4-Mini with Ollama

```
from hanzo import HanzoAI

client = HanzoAI()

# Zen4-Mini: fast, cost-efficient, good for high-throughput
    applications
response = client.chat.completions.create(
    model="zen4-mini",
```

```
    messages=[{"role": "user", "content": "Summarize this article in 3 bullet points."}],
    max_tokens=256
)
```

Listing 2: Zen4-Mini via Hanzo API

7 Safety and Alignment

Safety Metric	Zen-Eco	Zen4-Mini
Harmful content refusal	89.4%	95.8%
Over-refusal rate	12.1%	5.7%
Jailbreak resistance	82.3%	93.4%
TruthfulQA	58.7%	67.3%

Table 10: Safety Comparison: Zen-Eco vs. Zen4-Mini

Compact models have historically had weaker safety properties than larger models due to limited capacity for both capability and alignment objectives. Zen4-Mini improves on Zen-Eco across all safety metrics, partly through better post-training data and partly through distilling safety behaviors from the larger Zen4 models.

7.1 Limitations

- Complex multi-step mathematical reasoning is best handled by Zen4 or Zen4-Pro; Zen4-Mini's GSM8K score (81.2%) does not generalize to competition-level math.
- Long-context performance beyond 16K tokens degrades relative to full-context models, as the sliding-window attention limits long-range information propagation.
- The model is not recommended for applications where hallucination risk is critical; larger models in the Zen4 family have substantially better TruthfulQA scores.

8 Conclusion

Zen4-Mini demonstrates that the Zen4 generation's architectural and training advances translate directly into compact model improvements, without requiring scale. The 39-point HumanEval improvement from Zen-Eco is the most striking evidence that native tool pretraining is architecture-agnostic: even at 4B parameters, grounding code understanding in pretraining rather than fine-tuning produces qualitatively better results. With 1,200 tokens per second on consumer hardware, 2.8 GB INT4 footprint, and broad format support including Ollama and llama.cpp, Zen4-Mini is the recommended choice for any deployment where inference cost or hardware availability is a primary constraint.

Acknowledgments

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References

A Model Card

Field	Value
Model Name	Zen4-Mini
Version	v2026.01
Release Date	January 2026
Parameters	4.1B
Context Length	32,768 tokens
License	Apache 2.0
HuggingFace	huggingface.co/zenlm/zen4-mini-instruct
Ollama	ollama.com/library/zenlm/zen4-mini
Documentation	papers.zenlm.org/zen4-mini
Contact	research@hanzo.ai

Table 11: Zen4-Mini Model Card

B Quantization Quality Retention

Quantization	MMLU	HumanEval	GSM8K
BF16 (reference)	71.8%	74.3%	81.2%
INT8	71.6%	74.1%	81.0%
Q5_K_M	71.3%	73.8%	80.6%
Q4_K_M	70.9%	73.2%	79.8%
Q2_K	68.4%	69.7%	76.3%

Table 12: Benchmark Performance by Quantization Level

Zen4-Mini retains 99–100% of BF16 benchmark performance through Q4_K_M and 95–96% through Q2_K. The consistent retention across quantization levels reflects the architecture's low sensitivity to weight precision, partly due to RMSNorm (which avoids the weight-scale sensitivity issues of LayerNorm) and the symmetric distribution of weight values encouraged by the training recipe.